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Continental deposits, magnetostratigraphy and vertebrate paleontology, late Neogene of Eastern Spain

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Abstract

Eleven sections have been sampled for magnetostratigraphy in the fossil mammal-productive late Neogene basins of eastern Spain. The characteristic direction of magnetization was determined through progressive thermal and alternating field demagnetization. It was possible to correlate the magnetozones from the various sections to the GPTS as well as to the fossil MN zones within the sediment. The MN zone boundaries based on this correlation are as follows: MN 11/12≃8.1 my; MN 12/13≃6 my; MN 13/14≃4.9 my; MN 14/15=4.18 my; MN 15/16=3.58 my. New fossil lists are presented for the Fuente del Viso and Loma del Castillo faunas in the Cabriel Basin. © 1997 Elsevier Science B.V.

Keywords: magnetic stratigraphy; mammal evolution; biochronology; vertebrate paleontology

1. Introduction: Previous work

The Plains of La Mancha, near Albacete, and the nearby Teruel–Alfambra Basin in eastern Spain have been of interest to vertebrate paleontologists for more than 25 years. Fossils from this area were reported by Adrover (1968); the Venta del Moro vertebrate fauna was described several years later (Aguirre et al., 1973), and rodent faunas from the Teruel–Alfambra Basin were described by Van de Weerd (1976). Crusafont (1965) had proposed the Turolian Stage for the mammal sequence in the Teruel–Alfambra Basin, and Van de Weerd (1976) demonstrated a sequence of Vallesian, Turolian and Ruscinian rodents in deposits of that area. Mein et al. (1978) reported large and small mammals from the Jucar and Cabriel valleys, including a sequence of productive sites in the vicinity of Valdeganga, about 25 km northeast of Albacete.

The work reported below, correlating the deposits of the Jucar–Cabriel valleys and the Teruel–Alfambra Basin (Fig. 1), was initiated in 1980. Additional work was undertaken in 1989

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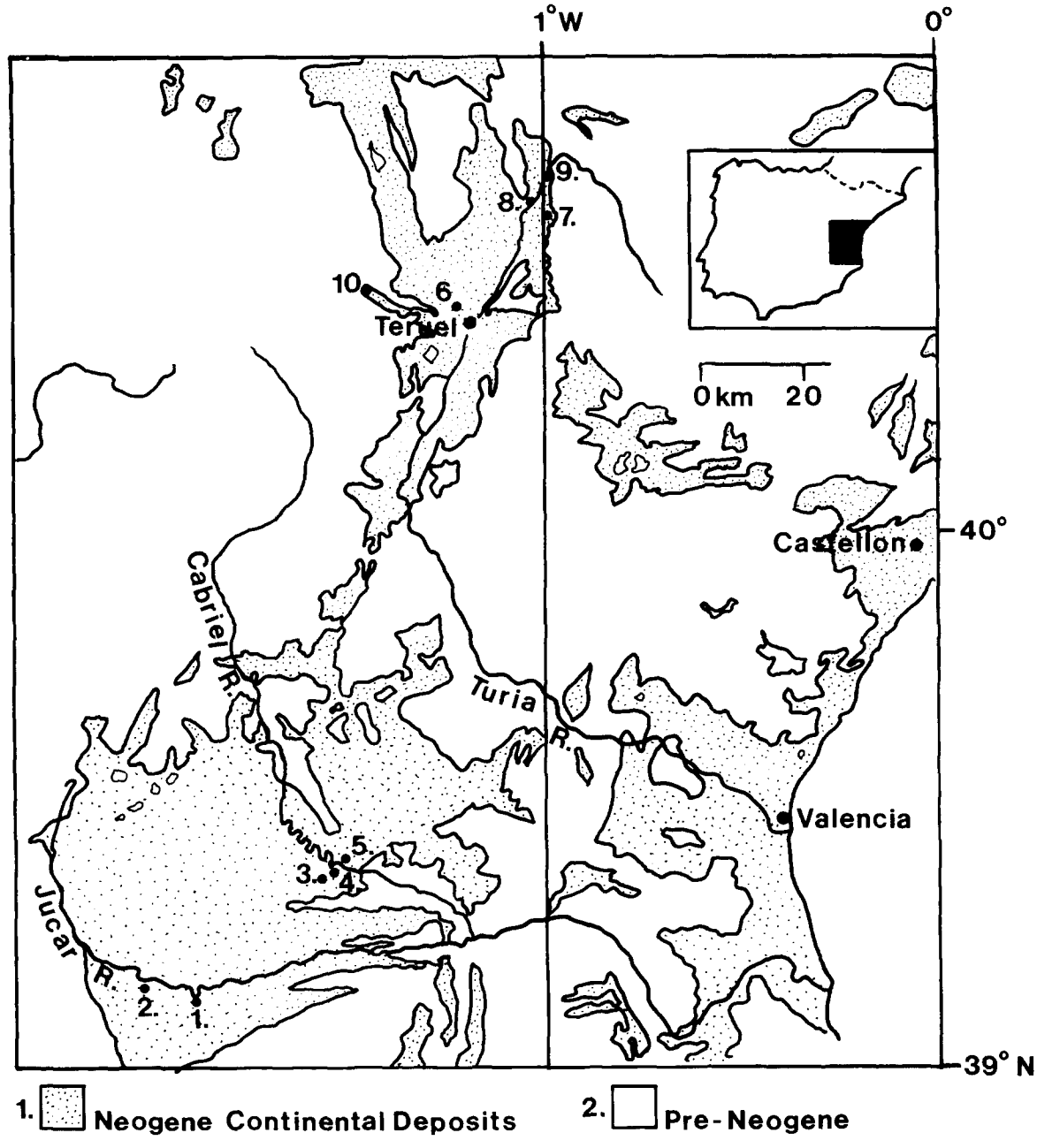


Fig. 1. Outcrop map of Neogene terrestrial sediments in eastern Spain. Sampling localities are: *1* = Valdeganga; *2* = Carrasco; *3* = Loma del Castillo; *4* = Cabriel Valley, South; *5* = Cabriel Valley, North; *6* = Concud Station; *7* = Escorihuela; *8* = Orrios; *9* = Villalba Alta and Villalba Alta Rios; *10* = Gea.

and 1990. Preliminary results were published in 1989 (Opdyke et al., 1989) and were presented at the 1990 meeting of the Regional Committee on Mediterranean Neogene Stratigraphy at Barcelona (Mein et al., 1990; Moissenet et al., 1990). The following results reflect a collaborative effort span-

ning more than 10 years; they were slow in maturation but should be of broad interest because they provide a chronologic framework for correlation of European late Neogene mammal faunas.

The prime objective of this collaborative project is to correlate securely placed mammal sites in long stratigraphic sequences by magnetic-reversal stratigraphy. Eastern Spain has proven an ideal location to accomplish our objective; fossils are relatively abundant, stratigraphic sequences are relatively long, and sediments are usually suitable for magnetic polarity studies. We have been able to place fossil sites, many of which are superposed in the same sequence, into 11 stratigraphic sections in eastern Spain. Collectively, the fossil sites span late Miocene, Pliocene and early Pleistocene time; their correlation with the geomagnetic polarity time scale provides a chronologic framework for partial calibration of MN zones 12 through 16.

Our sections can readily be identified by geographic area and chronologic interval, so we have prepared this report with three logical divisions: Jucar Valley, with emphasis on late Pliocene and Pleistocene faunas; Teruel–Alfambra Basin, with emphasis on Pliocene faunas; and the Cabriel Valley, with emphasis on late Miocene to Pliocene faunas.

2. Magnetostratigraphy

The sections chosen for study in all cases coincide with important vertebrate fossil localities. Attempts were made to select sections which were as long as possible in an attempt to sample a significant amount of time. Sections were taken preferentially in road cuts or dry washes in order to obtain unweathered samples. Sites were selected where possible in sediments of silt or finer grain size. Sandstones were generally avoided. It is desirable to place the sites at intervals of several meters; however, because of the variable nature of the lithologies in most of the sections the sites sampled were unevenly spaced in the sections.

At each site the surficial weathered material was removed with a pick mattock exposing an unweathered surface. The samples were oriented by first making a horizontal or vertical surface on the sample using a hand rasp and orienting the sample using a Brunton compass. The orientation was marked on the sample using a pencil. Three or more samples were taken from each site and returned to the laboratory where they were fashioned into 2.4 cm cubes. The study comprises samples from eleven sections in three sedimentary basins ranging in age from the late Miocene to late Pliocene.

3. Laboratory procedures

All samples were subjected to progressive thermal or alternating field demagnetization, with thermal demagnetization being the preferred treatment. The samples were measured in incremental steps at eleven or more intervals to temperatures as high as 670°C. The directions of magnetization were determined after analysis of orthogonal plots which allowed the determination of the various components of magnetization present in the samples. The direction of magnetization for each sample was determined by line fitting to the characteristic component of magnetization (Kirschvinck, 1980). As expected, the NRM intensity of samples ranges over several orders of magnitude; from lithologies that are barely measurable (e.g., the Alcala Limestone) to those that are strongly magnetic, such as the reddish siltstones of the Cabriel and Villalba Alta sections. The magnetic minerals which carry the magnetizations can be either magnetite, which unblocks by 580°C, or hematite, which unblocks at temperatures as high as 680°C, or in some cases goethite, which unblocks by 250°C.

The paleomagnetic results will be addressed beginning with the younger sequences in the Valdeganga and Teruel Basins. The sampling at Valdeganga was done on two occasions, the first in 1985 and the second in 1987. Resampling was necessary because of the very low intensity of magnetization of some samples. The exposure in the Valdeganga region has been subdivided into two formations (Mein et al., 1978): the Alcala Formation at the base, and the Valdeganga Formation above. The two sections sampled were

on both sides of the road between the village of Valdeganga and the Alcala River (Fig. 1).

About 10 km up-river from Valdeganga a second section was sampled in 1987 at the village of Carrasco. Thirteen samples were taken from 50 m of section. The Alcala Formation is present in the bottom 7 m of the Carrasco section. This formation is in turn overlain by the Valdeganga Formation. The lithologies of the formations at Carrasco are similar to that at Valdeganga; however, the lithology becomes more clastic along-strike toward Carrasco and, consequently, the percentage of limestone becomes less in the Carrasco section.

4. Magnetic properties

As might be expected from sequences with such varied lithologies, the magnetic properties of these sediments are very variable. One of the outstanding characteristics of these sediments is the fact that in some samples and sites the NRM is within the noise level of the magnetometer (5×10^{-5} AM). This is remarkable since these are the only sediments in my (NDO) experience in which the NRM is below the level of detection on a modern cryogenic magnetometer. Naturally enough this has led to difficulty in determining the magnetic polarity stratigraphy and is the basic reason for the resampling of the Valdeganga section.

As a result of this problem, it was decided to do a.f. demagnetization on all Valdeganga samples at steps of 2 mt to 20 mt. This was carried out on a commercially available a.f. demagnetizer by placing the samples in three orthogonal positions and then measuring them in a cryogenic magnetometer. All procedures were carried out in a double walled steel room in which the ambient field is reduced to 40 nannotesla. After a.f. demagnetization to 20 millitesla, the samples were heated to 250°C and cooled in a field-free space to ensure that the characteristic direction of magnetization was not being obscured by a high-coercivity low-blocking temperature mineral such as goethite.

Fig. 2a displays orthogonal plots from site 55, which is midway toward the top of the Valdeganga Formation in the section west of the road. These plots display the behavior of two samples in one site to a.f. and thermal demagnetization. In the case of sample 55a (not figured), the sample starts out weakly magnetized and by 10 millitesla the sample intensity has increased by a factor of two. The polarity is always reversed and the intensity of magnetization is the same as the NRM after demagnetization at 20 mt. The sample was then subjected to thermal demagnetization, which essentially did not effect a change until the temperature exceeded 450°C, when the intensity dropped below the noise level. Although weakly magnetized, the sample has a high coercivity and moderate blocking temperature. Fig. 2a shows the behavior of the second sample from this site. It can be seen that the NRM is one order of magnitude higher than the A sample. However, at 20 mt, it has demagnetized to the same level as the A sample and thereafter decays toward the origin. It is reversely magnetized. The direction remains reversed and the intensity does not change with thermal demagnetization at 250°C and above to 400°C. The C sample from the site exhibits totally different behaviour (Fig. 2b). It is two orders of magnitude stronger than the A sample and has a normal direction of magnetization which decays to the origin at 20 mt, thermal demagnetization does not change the polarity, which unblocks by 350°C. Great caution must be exercised in interpreting polarity at this locality since it is clear that complete remagnetization in the present field is possible in these sediments.

Fig. 2c shows the results of a.f. and thermal demagnetization from another site. In this sample a.f. demagnetization reveals a northerly directed normal component. Thermal demagnetization drops the intensity of magnetization by an order of magnitude and changes the inclination to a negative value and southerly declination, thereby fully reversing the polarity; apparently a mineral with low blocking temperature but high coercivity, such as goethite, is present in this sample.

The Fisher (Fisher, 1953) statistics from these different localities are presented in Table 1 for all class 1 sites (that is, the site yields a statistically significant site mean direction). Two of the localities, Cabriel and Villalba Alta, yield positive reversal tests (Fig. 3). The overall mean direction from

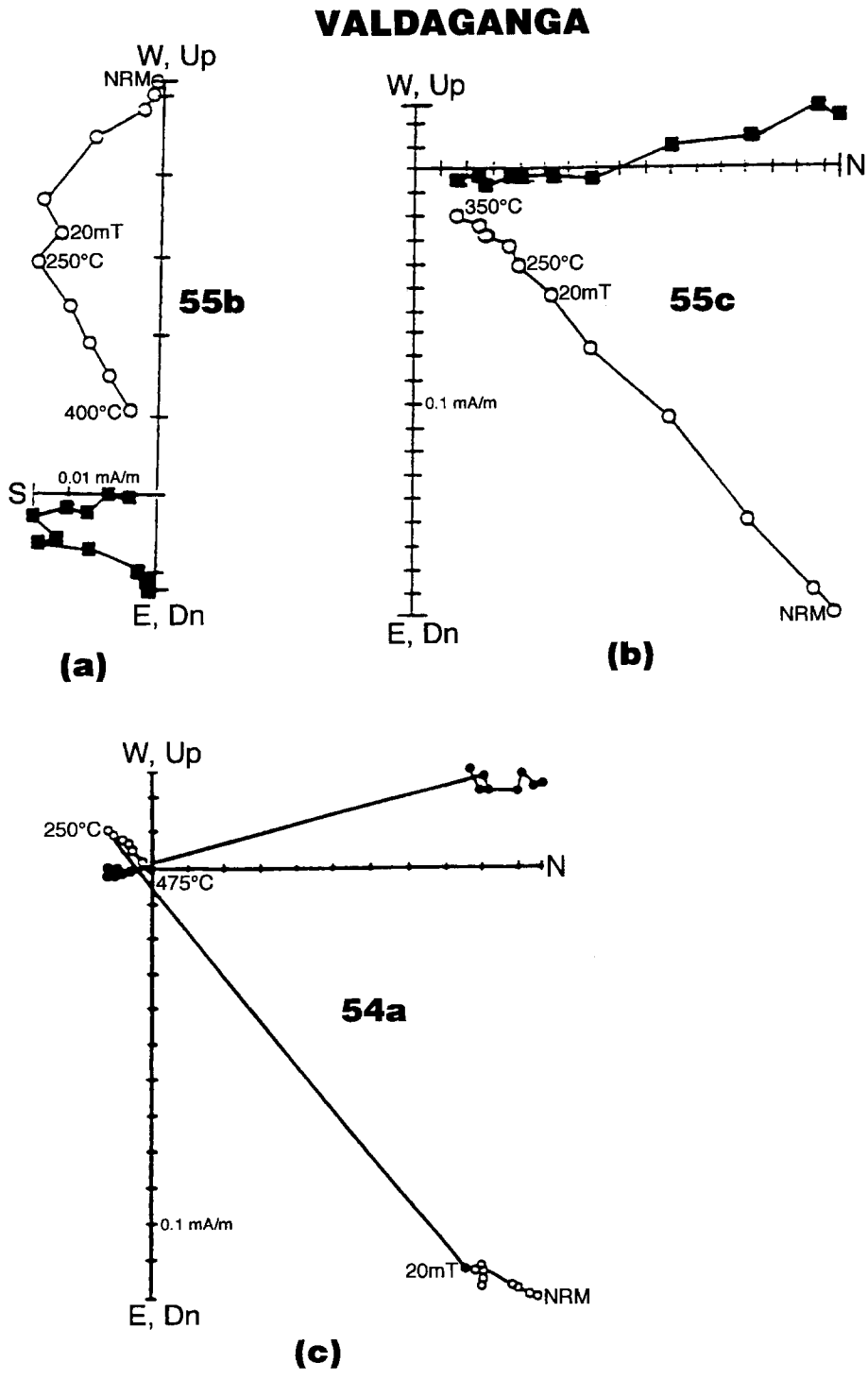


Fig. 2. Orthogonal plots of samples from Valdeganga. (a) and (b) are samples from the same site; sample 55c is completely remagnetized. (c) is from another site at Valdeganga, illustrating a dramatic shift after thermal demagnetization at 250°C and the ineffectiveness of a.f. demagnetization on these samples.

Table 1
Fisher parameters for paleomagnetic sampling locations. (Top) Reversed sites rotated to northern hemisphere

Location	N	R	K	α_{95}	DEC. (°)	INC. (°)	ΔDec
Cabriel	47	42.9	11.3	6.5	3.9	42.8	8.9
Carrasco	9	8.2	10.5	16.7	355.2	40.4	22.2
Concud	4	3.9	48.5	13.3	359.3	44.6	18.9
Gea	7	6.8	31.0	11.0	357.1	51.5	17.9
Orrios	4	3.9	22.8	19.6	350.9	56.6	37.5
Valdeganga 1	5	4.6	9.7	25.9	8.7	40.2	34.9
Valdeganga 2	2	2.0	37.5	42.0	349.0	51.7	[b]
Villalba Alta	13	12.2	14.2	11.4	0.5	54.5	19.9
Villanueva de la Jara	2	2.0	72.2	8.5	354.1	47.6	12.7
Location	N	R	K	α_{95}	DEC. (°)	INC. (°)	ΔDec
Cabriel [N]	31	28.9	14.1	7.1	5.8	44.2	9.9
Cabriel [R]	16	13.9	7.0	15.0	179.3	−41.1	20.1
Carrasco [N]	6	5.5	10.8	21.3	342.8	39.0	27.9
Carrasco [R]	3	2.9	28.4	23.6	199.3	−39.2	31.1
Concud [N]	3	3.0	62.2	15.8	355.5	41.6	21.4
Concud [R]	1[a]	2.8	10.6	39.9	193.3	−52.4	[b]
Gea [N]	5	4.9	34.3	13.3	357.7	52.3	22.1
Gea [R]	2	1.9	13.3	75.0	175.7	−49.5	[b]
Orrios[b]	4	3.9	22.8	19.6	350.9	56.6	37.5
Valdeganga 1 [N]	2	2.0	246.9	16.0	8.7	53.0	27.3
Valdeganga 1 [R]	3	3.0	40.3	19.7	179.4	−31.9	23.4
Valdeganga 2[c]	2	2.0	37.5	42.0	169.0	−51.7	[b]
Villalba Alta [N]	9	8.5	15.3	13.6	2.6	53.6	23.3
Villalba Alta [R]	4	3.7	9.7	31.1	175.2	−56.4	69.0
Villanueva[b]	2	2.0	872.2	8.5	354.1	47.6	12.7

[a]Only one class 1 reversed site. Used sample mean statistics, $N=3$.
[b]No reversed sites.
[c]No normal sites.
[N]=normal sites; [R]=reversed sites.

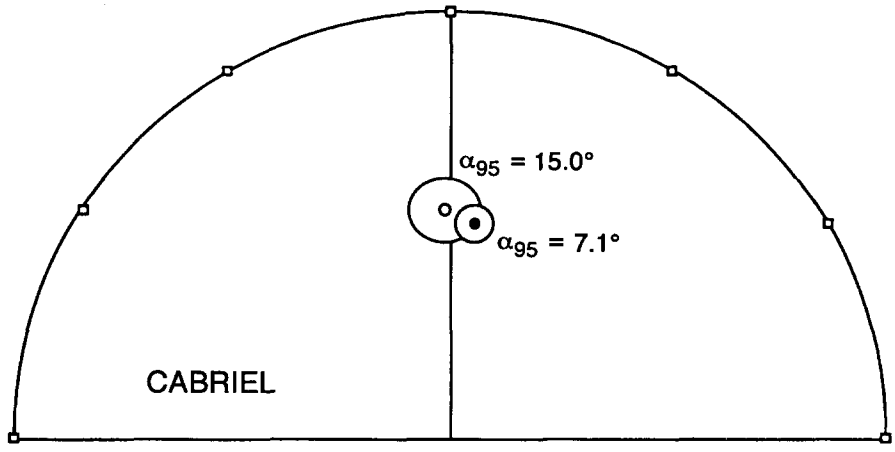
these localities is shallower than expected for the site latitude.

Samples from the Carrasco section have higher intensities of magnetization on average than those from Valdeganga. Fig. 4 gives the VGP latitudes for the sites from Valdeganga and Carrasco. All samples have been treated by thermal demagnetization in 50°C increments up to 300°C and 25°C increments between 300 and 550°C. As can be seen in Fig. 2, polarity determinations after cleaning procedures are not generally difficult to determine. However, the upper site at Carrasco is indeterminate as to polarity. At sites where VGPs could not be calculated, due to lack of data quality, the polarity is indicated by an open symbol.

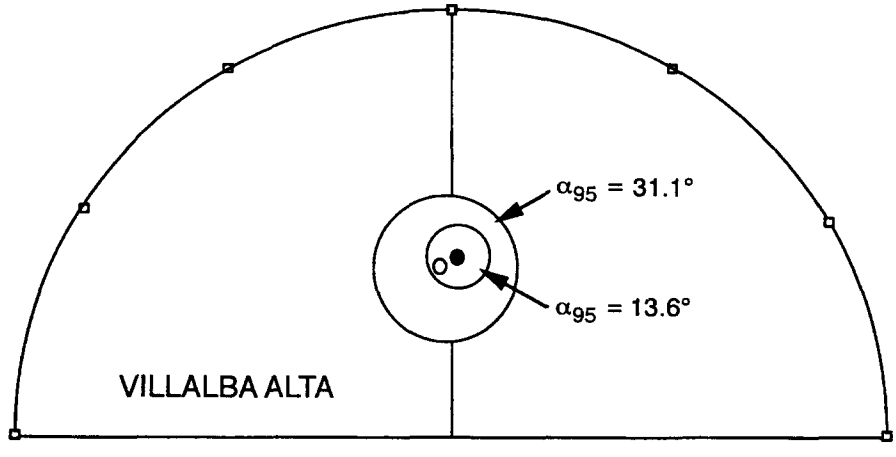
5. Magnetic stratigraphy

5.1. Valdeganga

The magnetic polarity sequence at Carrasco (Fig. 4) is a simple sequence with dominantly normal polarity in the Alcala Formation and reversed polarity in the Valdeganga Formation overlying it. The section at Valdeganga (Fig. 4) yields a more complex reversal sequence. The Alacala Formation yields a normal (N_1), reversed (R_1), normal (N_2) sequence. The overlying Valdeganga Formation is predominantly reversely magnetized; however, a normal magnetozone (N_3) is present. The fossils which are present in



Reversal test
- Normal sites (Dec. 5.8°, Inc. 44.2°)
- Reversed sites (Dec. 179.3°, Inc. -41.1°) (rotated to northern hemisphere)



Reversal test
- Normal sites (Dec. 2.6°, Inc. 53.6°)
- Reversed sites (Dec. 175.2°, Inc. -56.4°) (rotated to northern hemisphere)

Fig. 3. Reversal test at Villalba Alta and Cabriel.

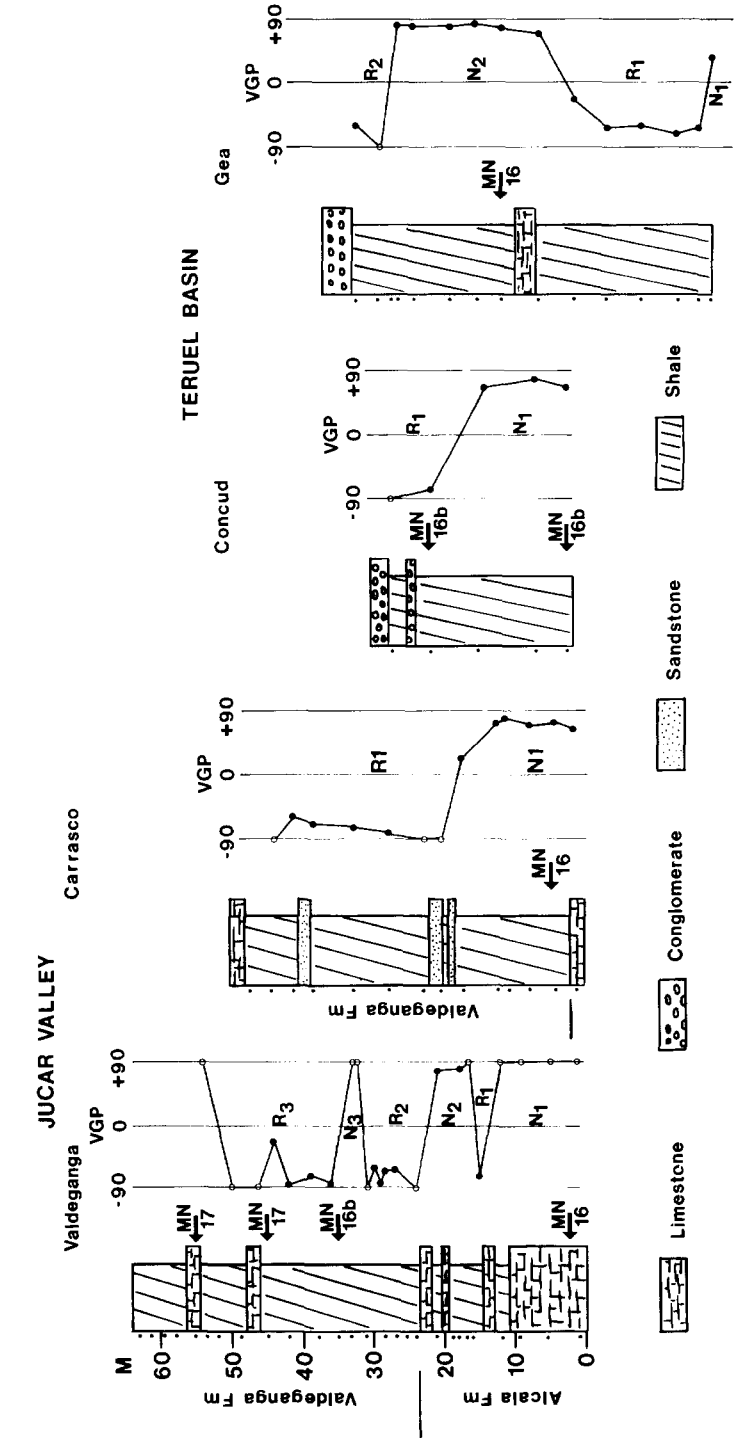


Fig. 4. Plots of VGP versus lithostratigraphic position of sites from late Pliocene–early Pleistocene of the Jucar Valley and the Teruel Basin. ○ = from sites where the polarity has been determined but the data quality does not permit a calculation of VGP latitude; ● = VGPs calculated from the mean of three samples. Position of fossil localities are given by the arrows.

these sections correlate to the Mammal Neogene (MN) zones 16a+b and MN 17, which are late Pliocene or Pleistocene in age (Table 2). A reasonable interpretation is that the base of the section at Carrasco is correlative with the Gauss Chron while the middle and upper part of the section is correlative with the Matuyama Chron. The interpretation favored for the Valdeganga section here is that the Kaena or Mammoth subchrons are represented by R_1 and the Olduvai subchron is represented by N_3. The fauna immediately above N_3 is placed in MN 17 zone, which is in agreement with an interpretation of N_3 as the Olduvai subchron.

Two other short sections have been collected from Spain which contain faunas of Pliocene–Pleistocene age. These sections were collected from the Teruel Basin further to the north of the sections previously described (Fig. 1). The first section in the Teruel Basin is the Concud Station section. It is composed predominantly of siltstone with minor conglomerates and with an organic layer at the base in which fossils occur. The section is only 30 m in thickness and five sites were collected. All samples were subjected to thermal demagnetization and VGP latitude plots are presented in Fig. 4. The orthogonal plots of these samples display simple univectorial decay to the origin with a low temperature overprint. The lower three sites of the section are normally magnetized and the upper two are reversed, although the upper site is heavily overprinted and yields widely scattered results. The original direction is believed to be of reversed polarity based on results obtained by thermal demagnetization.

The results from the short Concud Station section yield a reversal transition from normal to reversed. Unambiguous correlation to the GPTS is not possible and several possibilities exist. The transition could be associated with: (1) the normal to reversed transitions of the Kaena and Mammoth subchrons; (2) the Gauss–Matuyama transition; or (3) the upper transitions of either the Reunion or Olduvai subchrons within the Matuyama. The faunas belongs to MN Zone 16b and we therefore correlate the transition to the Gauss–Matuyama boundary; however, a correlation to the upper Olduvai subchron is also possible.

The last site to be discussed is the section at Gea (Fig. 4). This section is almost 60 m in length and consists of a series of siltstones with minor limestones, detrital limestones and calcareous siltstones. All samples were thermally demagnetized, usually in 14 steps to 680°C.

The data show at least two and maybe three magnetic transitions in the section (Fig. 4). The lowest site is normally magnetized; it is overlain by about 15 m of reversely magnetized sediments (R_1), which are overlain in turn by normally magnetized magnetozone (N_2), 30 m in length, the top of the sequence is again reversely magnetized (R_2). Given that the fossils (MN 16b) indicate an upper Pliocene–lower Pleistocene age, then the most probable correlation to the GPTS is that the normal zone (NZ) represents the late Gauss Chron. The fauna is similar to that at Carrasco; therefore, we correlate the normal magnetozone to the late Gauss Chron and the lower reversed zone (R_1) to the Kaena or Mammoth subchron. An alternative correlation is that N_2 represents the entire Gauss and that R_1 is the late Gilbert Chron. The site statistics for class 1 sites are given in Table 1. Stratigraphic ordering of the faunas, including records of important taxa, are given in Fig. 10.

5.2. *Teruel basin (and younger Cabriel basin)*

The next sections to be considered are those from the Teruel and Cabriel Basins which yield faunas of MN 14, MN 15 and MN 16 age at Villalba Alta, Villalba Alta Rio, Orrios, Escorihuela and Loma del Castillo, located in Figs. 1 and 6 (see faunal data in Table 2). Excellent paleomagnetic results were obtained from the Villalba Alta and Escorihuela sections; the amount of calcite increases in sediments of the Orrios section and the intensity of magnetization drops, dramatically affecting the quality of the paleomagnetic results. However, it was possible to determine the polarities of most sites unambiguously. Representative orthogonal plots from the four sections are given in Fig. 5, and the Fisher statistics are presented in Table 1. The composite section at Villalba Alta (Fig. 6) yields seven magnetozones and reveals a pattern that can be matched to the

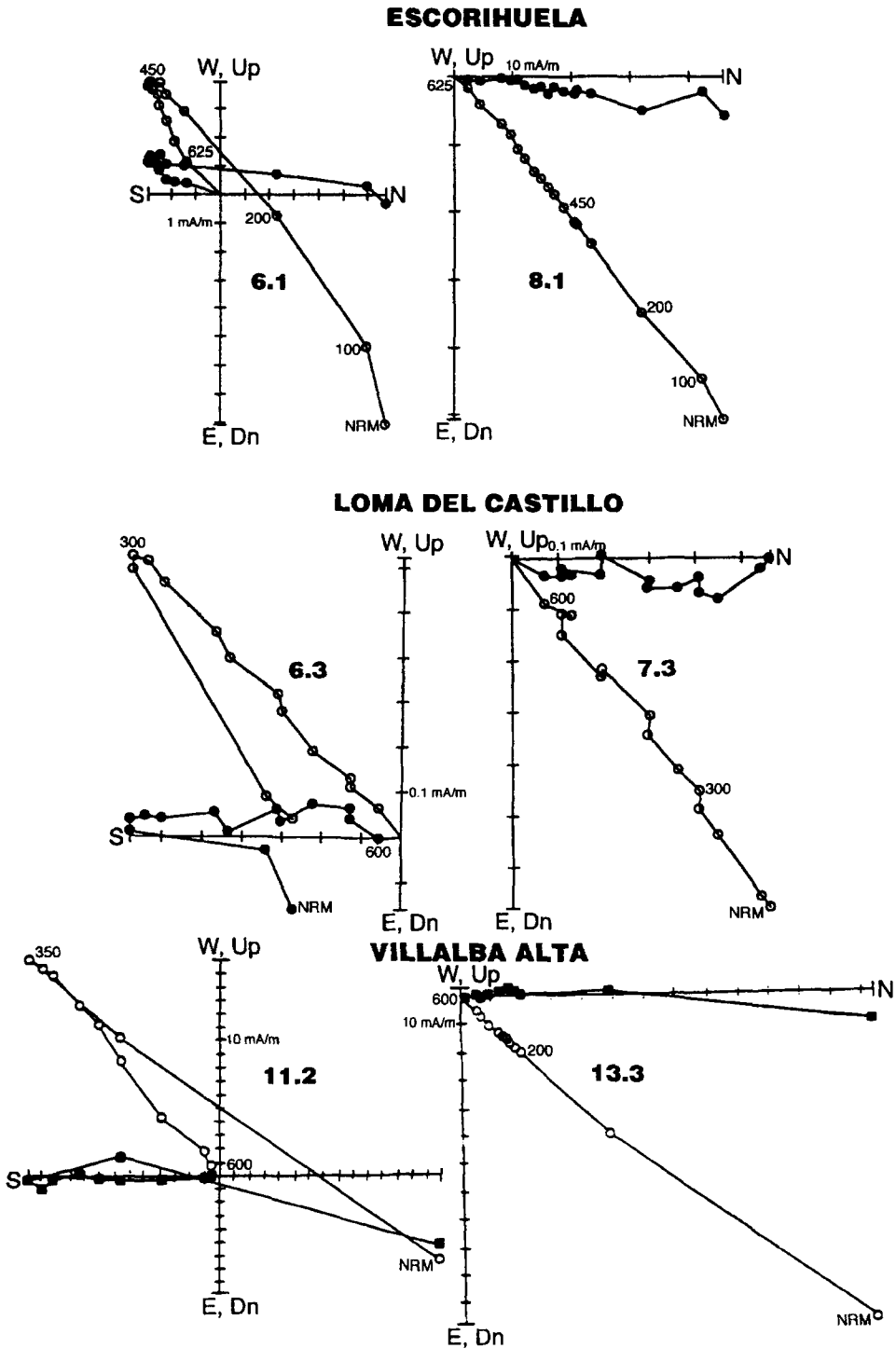


Fig. 5. Orthogonal plots of samples from Escorihuela, Loma del Castillo and Villalba Alta.

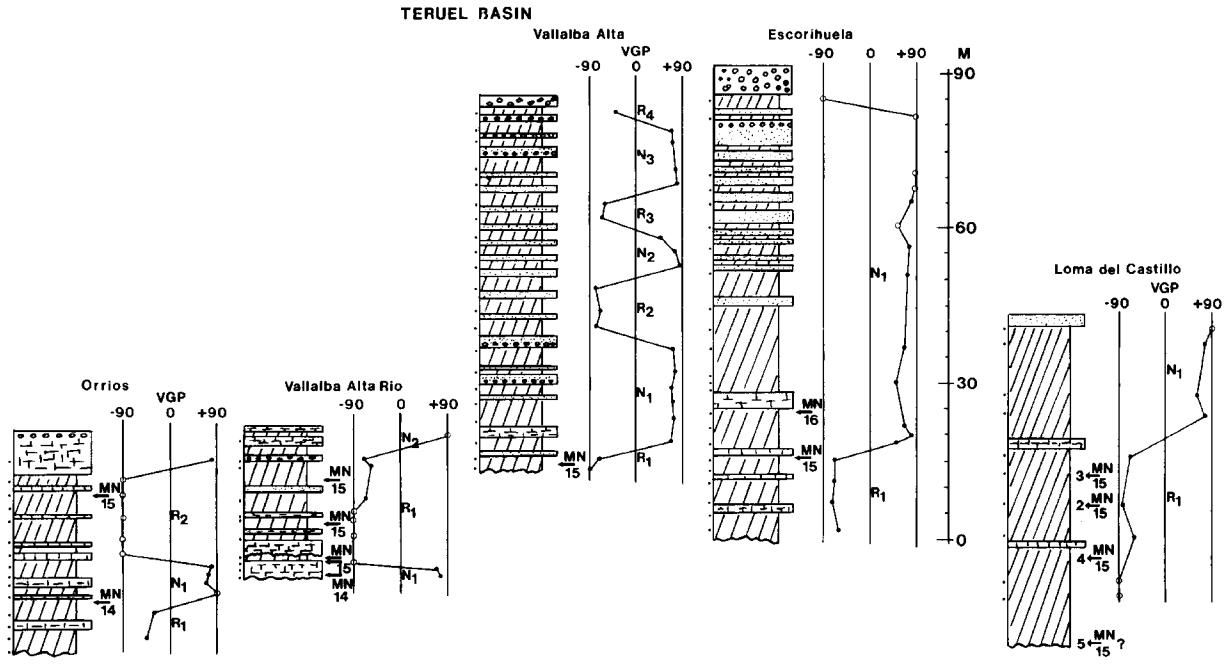


Fig. 6. Plots of VGP versus lithostratigraphic position for the sections at Orrios, Villalba Alta Rio, Villalba Alta, Escorihuela and Loma del Castillo. Symbols as in Fig. 3.

magnetic polarity time-scale from the early Matuyama, through the Gauss, to the late Gilbert subchron (Fig. 9). The pattern at Orrios, R_1 to N_2, may be correlated laterally to the Villalba Alta Rio section and, in turn, to the Villalba Alta section, with magnetozone N_1 in the Orrios and Villalba Alto Rio sections probably correlative with the Cochiti subchron.

The important section at Escorihuela is separated from those in the Orrios, Villalba Alta area by 5 km. This sequence proved to be strongly magnetized and yielded excellent results in the basal two thirds of the section. The samples often unblocked at very high temperature (680°C), indicating the presence of hematite as the dominant magnetic carrier. The bottom 35 m of the section is reversely magnetized and the remainder of the section is of normal polarity, with the exception of the top site which is interpreted as having reversed polarity. Two fossil localities are located in this section; the stratigraphically lowest site (Escorihuela B) is MN 15 zone, in sediments that are reversely magnetized. The upper locality (Escorihuela A) (MN 16) is in sediments of normal polarity near the base of magnetozone N_1. We therefore correlate the R_1–N_1 transition in this sequence to the Gilbert/Gauss boundary. The magnetic stratigraphy of these sections is presented in Fig. 6. The section studied at Loma del Castillo is situated in the Cabriel Basin, along-strike from the long section from the south bank of the Cabriel River described below, but higher stratigraphically than the top of the Cabriel section. It has yielded good results, demonstrating that two magnetozones are present, R_1 and N_1. Four fossil sites of MN 15 age are present in magnetozone R_1 of the Loma del Castillo section. Magnetozone R_1 is interpreted to represent late Gilbert age with the overlying normal zone representing the early Gauss Chron. The lowest fossil site (below the magnetically sampled section) may yield an MN 14 fauna.

5.3. *Older Cabriel basin*

In 1989, Opdyke et al. reported magnetostratigraphic results from late Miocene sediments in the

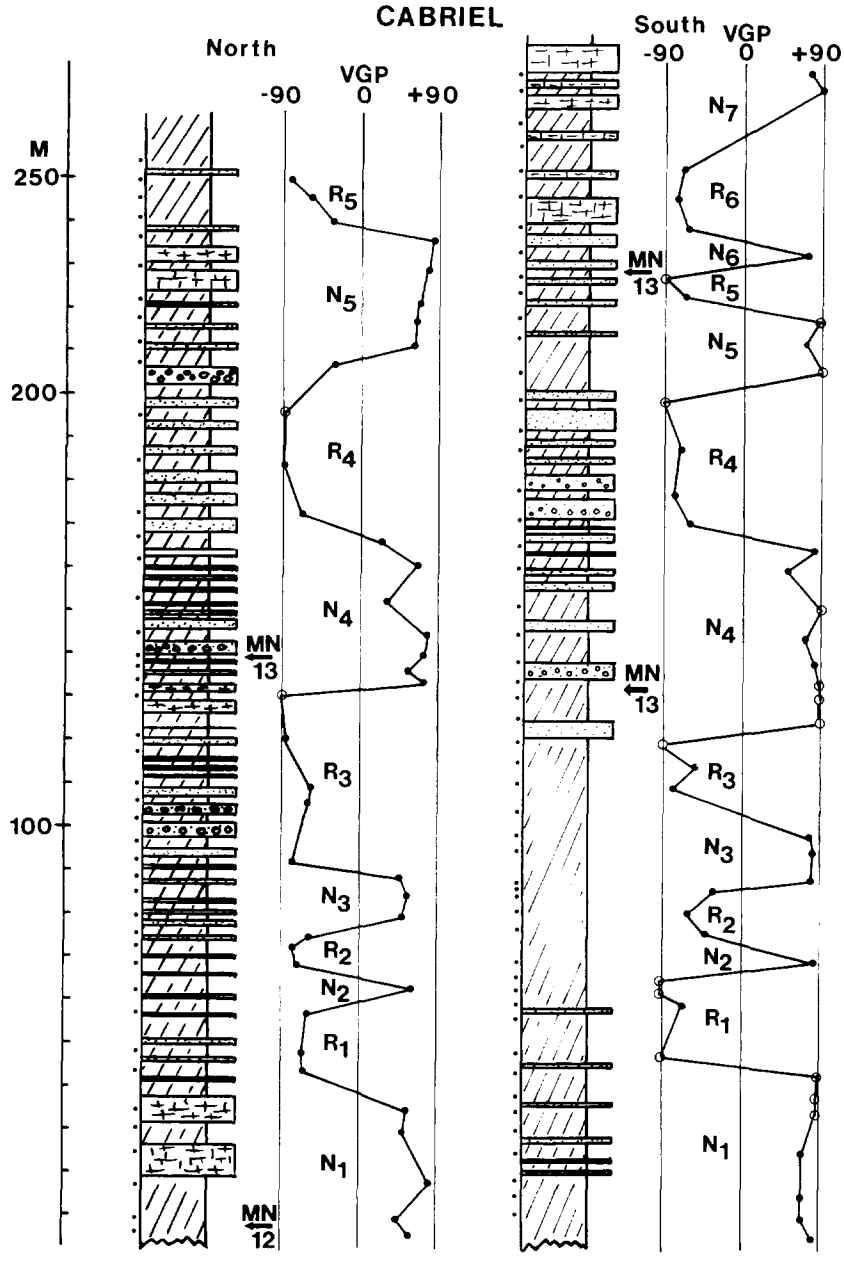


Fig. 7. Plots of VGP versus lithostratigraphic position for the sections at Cabriel North and South. Symbols as in Fig. 3.

Cabriel Basin that yielded faunas from MN 11 to MN 13 age. The sediments in that study were sampled along the road north from the Rio Cabriel. An excellent magnetic stratigraphy was determined from this section which were interpreted to range in age from Chron 7 to early Gilbert Chron (late Miocene). An important fossil site (Fuente del Viso) was subsequently found in the upper part of the sedimentary sequence on the southern shoulder of the Cabriel Valley. The fossils from this site were later determined youngest MN 13 in age, presumably very close to the MN

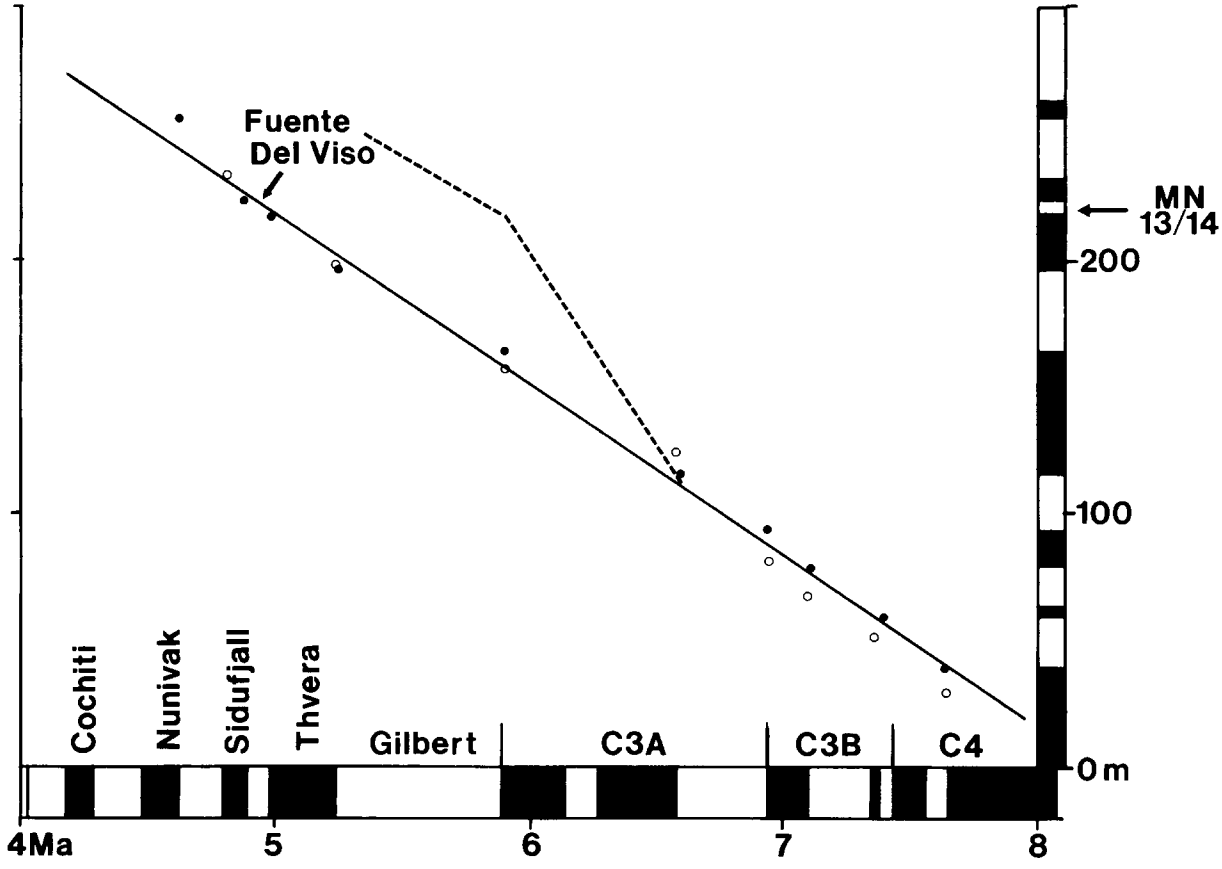


Fig. 8. Plot of magnetic stratigraphy from the south flank (dots) of the Cabriel Valley to the MPTS of Cande and Kent (1995). The position of the same transition from the north flank is indicated by the circles.

13–MN 14 boundary. The section exposed on the road south of the Cabriel River extends higher in the sequence than the sequence reported on the north (left bank) in 1989; therefore, a section was sampled on the south bank along the road south of Villaroya. The section on the left bank of the river was remeasured and the length of the section was found to have been underestimated by about 25 m in 1989.

The paleomagnetic samples again gave satisfactory results; representative orthogonal plots were given in 1989. The plot of the VGPs against the stratigraphic position is shown in Fig. 7. At some sites polarity could be determined but the data was inadequate to calculate VGPs. The boundary between magnetozones are placed using the entire data set. Eleven magnetozones were determined which can easily be correlated from one side of the river valley to the other (see Fig. 7).

Several different possible correlations of this stratigraphy to the GPTS were examined. The original (1989) correlation favored equating magnetozones N_4 through N_5 on the north bank to Chron C3A (Chron 5). The expansion of the reversed magnetozone R_4 in the profile from the south bank and the pattern observed in N_5 through N_7 in the southern section suggested that a better fit of the pattern could be made if these magnetozones were correlated to the Thvera, Sidufjall and Nunavak subchrons. Therefore a diagram comparing the observed magnetostratigraphic sequence with the GPTS was plotted in a time vs. stratigraphic position diagram and is presented in Fig. 8. The magnetostratigraphic

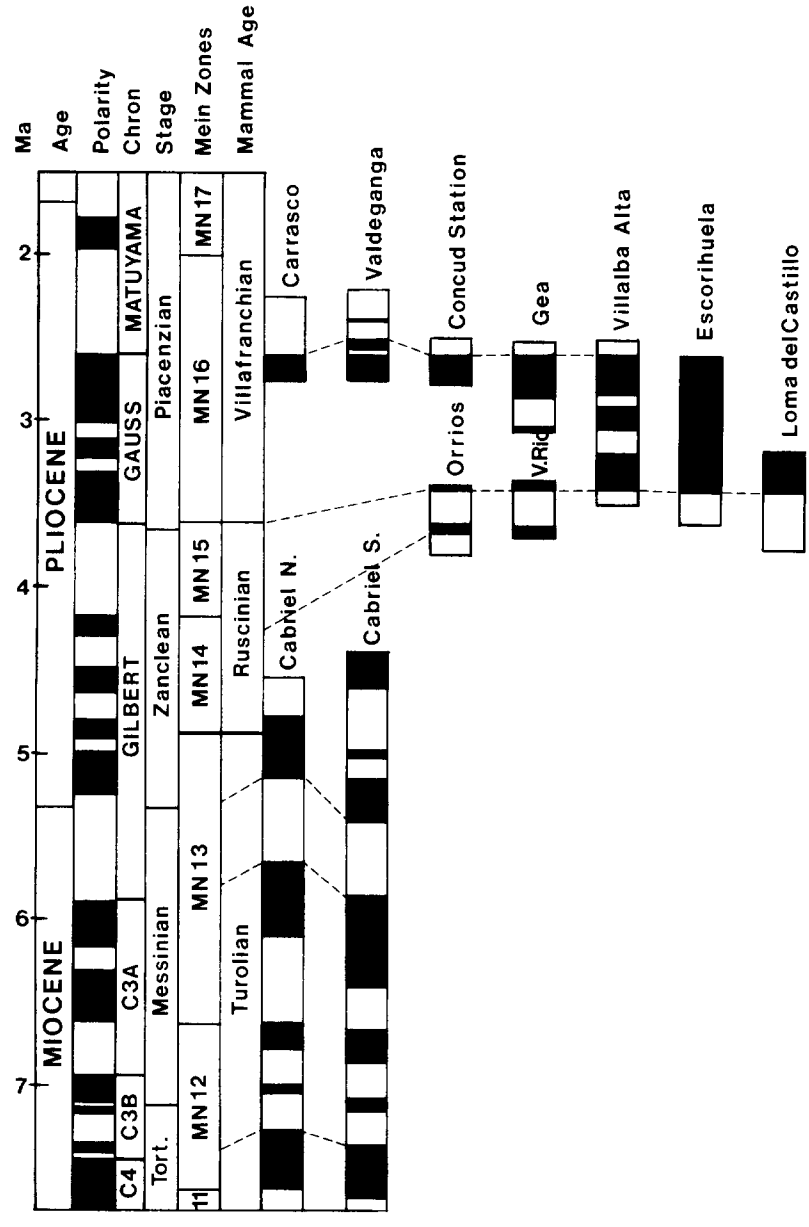


Fig. 9. Plot of the position of the MN zones from MN 11 through MN 17 in the various sections studied and their position with respect to the MPTS, the stages of the late Miocene through Pleistocene and to the European land mammal ages.

pattern is seen to correlate well for the interval from Chron C4n.1n to the Nunivak subchron (Chron C3n.2n) of the Gilbert Chron. An alternative correlation, indicated by dashed lines in Fig. 8, would correlate magnetozone N_4, N_5 and N_6 to C3A. This correlation to the time scale requires a large change in sedimentation rate in the sedimentary basin which is not reflected in the lithologies. Therefore the correlation of the top of the Cabriel sedimentary sequence to the middle Gilbert Chron is favored. This correlation is shown in Fig. 9 and implies that the Turolian–Ruscinian

boundary is not coincident with the Miocene–Pliocene boundary as has generally been assumed; instead, the Turolian/Ruscinian boundary (e.g., MN 13/14) is lower Pliocene in age. The Fuente del Viso fossil locality (latest MN 13), which occurs between N_5 and N_6, would therefore fall between the Sidufjall (Chron C3n.3n) and Thvera (Chron C3n.4n) subchrons at 4.93 Ma. The rate of sedimentation in the basin would be about 6.8 cm/1000 yr.

The important fossil locality of Venta del Moro lies above a transition from normal to reversed polarity and was reported by us in 1989. The correlation of this reversal to the GPTS is uncertain, but a position above the N_4/R_4 transition was favored at that time. This position is still favored because the fauna at Venta del Moro is considered primitive relative to that of Fuente del Viso. The age of Venta del Moro is now believed to be about 5.8 my, situated in the lower Gilbert Chron based on the correlation given here.

Another section has been sampled (but is not presented) at Barranca de Cuevas in an attempt to bridge the gap between MN 13 and MN 14. The magnetic stratigraphy of this section is not very clear and the data quality is variable. However, the base of the section is normally magnetized and this is followed by a 40 m reversed interval. A single normal polarity site is present at the fossil site which produced fossils considered earliest MN 14 zone. The fossils from Barranca de Cuevas cannot be placed unambiguously in the GPTS at this time; however, a position for this site within one of the Gilbert subchrons is feasible.

6. Vertebrate paleontology

6.1. Fossil sites and collecting methods

Cenozoic sediments in eastern Spain have proven very productive for small mammals. Daams and Freudenthal (1988) summarized sequences of small mammal faunas collected from early to middle Miocene sediments of the Teruel Basin from 1975 through 1986. Our work was concentrated mostly to the south of the sequences reported by Daams and Freudenthal and collected mostly after their work. Our work complements their studies and extends into younger sediments. The basic techniques of fossil recovery are comparable, however, to those described at some length by Daams and Freudenthal (1988). In total, 20 sites were sampled for small mammals in the areas we studied and 18 of these have proven productive. The fossil sites were traced into well exposed and relatively continuous sections from which paleomagnetic samples were collected to yield a magnetic reversal sequence. The resulting biostratigraphic–magnetic polarity sequence can then be calibrated if the magnetic sequence can be correlated with the magnetic polarity time scale.

6.2. Mammal Neogene (MN) zones

Mammal Neogene (MN) zones were proposed by P. Mein in 1974 and formally presented a year later at the Bratislava meeting of the Regional Committee on Mediterranean Neogene Stratigraphy (e.g., RCMNS; Mein, 1975). As proposed, and later revised (Mein, 1989), these MN zones provide a framework for correlation of European faunas, based primarily on the appearance (and extinction) of well known and widespread mammals, both large and small. They have proven very effective and the revised mammal zonation (Mein, 1989) is now one of the more reliable paleontologic–chronologic frameworks for late Cenozoic history of Europe. This work is directed toward the chronologic resolution and calibration of MN zones by correlating representative fossil sites with the magnetic polarity time scale. Our focus in this study is on MN zones 12 through 16, spanning late Miocene, Pliocene, and early Pleistocene faunas of Europe.

6.3. A composite section

Fig. 9 gives the position of the magnetozones determined in this study, correlated to the magnetic polarity time scale for late Miocene and Pliocene time. The composite sequence correlates the magnetic sequences from the various basins ranked according to age. The placement of the fossil

Biochronology of the late Neogene of Eastern Spain

		APPEARANCE	JUCAR FAUNAL SITES	TERUEL FAUNAL SITES		LAST OCCURRENCE
MN 17	zone of *Mimomys pliocaenicus*		Valdeganga 2, 3, 6			
			Valdeganga 7, 10			
MN 16	zone of *Kislangia*	*Equus, Mammothus Kislangia*	Rincon 1 Valdeganga 1, 11, 4, 8 Rincon 2, Canada Real	Concud Estacion 2 Concud Estacion 1 Concud Pueblo Gea 1	Orrios 8	
				Gea 0		*Ruscinomys Eliomys, Blancomys, Trilophomys*
	zone of *M. gracilis + M. hajnackensis*		El Carrasco	Teruel - Transformadores		
			La Marmota	Escorihuela	Sarrion 1	
			Valdeganga 15	Escorihuela A		
		Cseria	Valdeganga 14	Escorihuela C	Orrios 5	
			Loma del Castillo 3	Poblado Iberico		*Dolomys*
		M. occitanus, Sorex	Loma del Castillo 2	Villalba Alta 5 Escorihuela B		*Pliopetaurista*
	zone of *Dolomys*				Orrios 7	
						Paraethomys, Galerix
MN 15		*Dolomys, Blarinoides*		Teruel - Cementerio Villalba Alta 4 El Rajo	Orrios 3 Orrios 4	
				Villalba Alta 1		*Atlantoxerus*
	zone of *Mimomys archaiques*		Loma del Castillo 4		Sarrion 2	
		M. davakosi		Arquillo 3 Teruel dessio		*Paraet. abaigari*
			Loma del Castillo 5	Villalba Alta 3	Lomas de Casares 1	
		M. vandermeuleni *Mimomys, Pliopetaurista*	Loma del Castillo 1	Villalba Alta Rio 2	Gloria 3 Gloria 2	
		Blancomys neglectus		Villalba Alta Rio 1	Aldehuela	
	zone of *Trilophomys*			Teruel Plaza de Toros		
				Teruel - Juderia		
				Orrios 1	Celadas 6	
		Trilophomys			Celadas 12	*Promimomys*
					Celadas 9	
MN 14	zone of 2 *Paraethomys* (+ *Promimomys*)	*Promimomys, Castillomys, Protatera*		Gloria 4	Cerro Gordo	*Protatera*
					Celadas 5	*Eozapus*
		Paraethomys abuigeri			Celadas 14	
						Amblycoptus, Celadensia
	zone of *Celadensia*			Peralejos E	Celadas 4	
				Lomas de Casares 2		
		Celadensia, Eozapus, Micromys, Rhagapodemus		Lomas de Casares 3	Celadas 8 Celadas 3	
			Fuente del Viso	Arquillo 4		*Blancomys sanzi*
		Paraethomys			Celadas 2	
				Arquillo 1		*Kowalskia*
MN 13	zone of *Stephanomys ramblensis*			Gloria 5 Valdecebro 6	Villastar	
		Blancomys, Hystrix		Valdecebro 3		
				Masada del Valle 7		*Karnimata*
				Gloria 6	Velilla 1	
		Stephanomys, Apodemus, Karnimata			Tramacastiel	
MN 12	zone of *Parapodemus gaudryi*					
		Parapodemus barbarae *Occitanomys adroveri* *Eliomys truei*	Fuente Podrida			
MN 11	zone of *Parapodemus lugdunensis*	*Parapodemus lugdunensis*	Balneario			

Fig. 10. Biochronology of the late Neogene of eastern Spain.

Faunistic Evolution in the Valdeganga Area (Spain)

MN	JÚCAR LOCALITIES	EVENTS ⌊First appearance ⌈Last appearance	IMPORTANT TAXA
17	Valdegangag 2, 3, 6 Valdeganga 7 and 10	 ⌊Eucladoceros	*Equus stenonis* 2 *Kislangia* cf. *rex* *Mimomys* cf. *medasensis* *Mimomys* cf. *reidi* *Stephanomys balcellsi* *Castillomys crusafonti* ssp. *Apodemus dominans* *Micromys* cf. *minutus*
	Rincón 1, Valdeganga 1 and 11	⌊Equus	*Equus stenonis* 1 *Kislangia* cf. *rex* *Mimomys realensis* *Mimomys* cf. *reidi*
16	Cañada Real, Valdeganga 4, 8 and 16b Rincón 2, Valdeganga 9, 9b and 16a	 (Hipparion ?) ⌊Micromys, Mammuthus	*Mammuthus gromovae* *Sus strozzi* *Kislangia ischus* *Mimomys realensis* *Stephanomys* cf. *balcellsi* *Castillomys crusafonti*
	El Carrasco La Marmota Valdeganga 15 Valdeganga 14	⌊Kislangia, Cseria	*Kislangia ischus* *Cseria gracilis* *Stephanomys minor* *Castillomys crusafonti* *Apodemus dominans*
15	Cuesta de Mahora, Abengibre, La Recueja	⌈Dolomys Occitanomys ⌊Dolomys	*Dolomys adroveri* *Stephanomys* cf. *donnezani* *Occitanomys brailloni* *Castillomys crusafonti* *Rhagapodemus* sp. *Apodemus dominans* *Muscardinus pliocaenicus* *Eliomys intermedius*

Fig. 11. Plot of the mammal successsion near Valdeganga and surrounding area in the Jucar Valley

localities and MN zones will be discussed in order of decreasing age.

In our 1989 paper we pointed out that the fossil locality at Fuente Podrida at the base of the Cabriel section contains fossils of MN 12 age (Mein et al., 1978), including *Parapodemus barbarae*, *Occitanomys adroveri* and *Eliomys truci*. Balneario, another small undescribed fossil locality, 300 m to the east and about 10 m below the site at Fuente Podrida, contains *Parapodemus lugdunensis*, which is a guide fossil for MN 11. The boundary between these two localities must determine the MN 11/12 boundary. The boundary would fall in Chron 4N, at about 8.1 my according to the Cande and Kent time scale (Cande and Kent, 1995).

The MN 12/13 boundary is not well constrained by this study. Undescribed MN 13 faunas occur in the Cabriel sections within basal magnetozone N_4. The boundary between MN 12 and MN 13 must be older than this. We arbitrarily place the boundary below the base of Chron C3An.2n (= magnetozone N4) at an age of about 6.7 my.

The important Fuente del Viso fossil locality is located between magnetozones N_5 and N_6 in the Cabriel southern section. The Fuente del Viso faunal list is as follows:

Insectivora	*Dibolia dekkersi*	
	Episoriculus gibberodon	
Lagomorpha	*Prolagus michauxi*	
Rodentia	*Ruscinomys lasallci*	*Apodemus gorafensis*
	Blancomys sanzi	*Eliomys intermedius*
	Cricetus barrieri	
	Stephanomys medius	
	Occitanomys sp.	
	Paraethomys anomalus	

It is considered latest MN 13 based on the stage of evolution. The correlation to the time scale favored here is that this fauna is best correlated to the interval between the Sidufjall and Thvera subchrons (e.g., Chrons C3n.3n and C3n.4n) of the Gilbert Chron and is given an age of about 4.93 my. The MN 13/14 boundary is therefore positioned slightly younger than this, at 4.9 my, at the base of the Sidufjall subchron (Chron C3n.3n).

This position for the MN 13/14 boundary is younger than its traditional position, correlated to the marine Miocene–Pliocene boundary (5.3 my), which corresponds to the refilling of the Mediterranean at the end of Messinian Stage. As far as the authors are aware there is no evidence that would make the correlation presented here impossible to support. It would seem that the significant change in terrestrial faunas was about 400,000 years after the Mediterranean refilled.

The next faunal change between MN 14 and MN 15 is represented in the Teruel basin at Orrios and Villalba Alta Rio. Late MN 14 localities are present in magnetozone N_1 in these sections. The Orrios 1 site and Villalba Alta Rio 1 site both contain *Trilophomys* cf. *castroi*, a guide fossil for the latest MN 14 mammal subzone. These fossil zones are succeeded by fossil sites which record the first appearance of *Mimomys*. The fauna from sites in the Villalba Alta Rio section (e.g., Villalba Alta Rio 2, 3, 4, and 5) contain fossils spanning most of MN 15. The boundary between MN 14 and 15 therefore occurs at the top of magnetozone N_1 in this section, which we correlate to the Cochiti subchron (Chron C3n.1n). We therefore place the MN 14/15 boundary at the top of the Cochiti subchron at 4.18 my. All the MN 15 faunas that have been examined in this study fall in reversely magnetized sediments, which we correlate to the late Gilbert Chron at Orrios, Villalba Alta Rio, Villalba Alta, Escorihuela and Loma del Castillo (see Fig. 6).

The boundary between MN 15 and MN 16 has been established at Escorihuela where an MN 15 fauna (Escorihuela B) containing *Dolomys adroveri* is stratigraphically overlain (an interval of 10 m) by two sites having MN 16 faunas (e.g., Escorihuela A and C) with *Cseria gracilis* and *Mimomys hajnackensis*. The magnetic field reverses to normal polarity just above the MN 15 (Escorihuela B) site and the MN 16 sites are from normally magnetized sediment. The faunal change is therefore very close to the Gilbert/Gauss boundary. We have plotted them as synchronous, which yields an age of 3.58 my for the MN 15/16

boundary. This boundary is usually considered correlative with the Ruscinian–Villafranchian/Villanyan boundary and is therefore of considerable importance. A recent study by Aguirre et al. (1995) places MN 15 faunas of the *Dolomys* subzone from reversely magnetized sediments in a section located in the province of Cádiz, southern Spain. This is in agreement with the results reported here. Aguirre et al. extended the top of the MN 15 zone into the early Gauss, although the data that they presented did not require that interpretation. The MN 15 faunas presented here all fall in the late Gilbert (five sections) in reversed sediments. The MN 15 sites at Loma del Castillo have not been reported previously. Their stratigraphic position is indicated in Fig. 6; faunas from the Loma del Castillo fossil sites are listed below. Loma del Castillo site 1 is 10 m below the base of our measured section; it may be MN 14 in age.

Loma del Castillo 1	*Cricetus barrieri*
	Apodemus dominans
	Apodemus gorafensis
	Paraethomys meini
	Occitanomys brailloni
Loma del Castillo 5	Arvicolid fragments
	Paraethomys meini
	Occitanomys alcalai
	Galerix sp.
Loma del Castillo 4	Arvicolid fragments
Cricetus barrieri	
	Paraethomys meini
Loma del Castillo 2	*Mimomys occitanus*
	Stephanomys fragments
Loma del Castillo 3	
Paenelimoecus pannonicus	
	Prolagus michauxi
	Dolomys adroveri
	Stephanomys margaritae
	Apodemus dominans
	Eliomys intermedius

MN 16 faunas occur in both normal and reversely polarized sediment, which we interpret to be equivalent to the Gauss and lower Matuyama chrons. The boundary between MN 16 and MN 17 faunas occurs at Valdeganga. The MN 17 faunas occur just above a short magnetozone that we correlate to the Olduvai subchron (Chron C2n). We therefore believe that the MN 16/17 boundary is younger than the Olduvai subchron and close to the Pliocene/Pleistocene boundary as determined in Italy (Zijderveld et al., 1991). Faunas of MN 17 age from central France (e.g., Rocca Neyra) have radiometric dates that would place them younger than the Olduvai subchron.

Fig. 10 summarizes the biochronology of our work in the Cabriel, Jucar and Teruel basins. Important fossil sites are ordered relative to their stratigraphic and paleomagnetic position, and placed into the MN zonation.

Table 2 illustrates the faunal change from MN 15 through MN 17 in the Jucar–Valdeganga area. Small mammals are well represented in the collections from that area made by P. Mein and E. Moissenet for this project. MN 16 and 17 can be ordered according to the sequence shown in Table 2.

7. Sedimentation

The sediments sampled in this study span several important regional and global environmental changes. The Messinian salinity crisis is spanned by the sediments of the Cabriel Basin; specifically, magnetozones N_3 to the top of R_4 in the Cabriel sequence. The sediments in this interval are predominantly clastic with brown color, although minor occurrences of gypsiferous lithology are present. The fresh water limestones in the Cabriel section, however, occur prior to or following the Messinian. These same fresh water limestones (the Alcala Formation) develop a thick (160 m) sequence of limestones in the Jucar River valley. The base of the Jucar sequence cannot be dated magnetostratigraphically; however, a micromammal site of MN 13 age occurs at Tolosa; it may be Messinian or post Messinian in age. The fresh water limestone deposition continues in the Jucar Valley to the top of the Gauss Chron and so spans most of the lower and middle Pliocene. These large lakes represented by the fresh water limestone deposits may reflect an increase in precipitation which occurred during and after the filling of the Mediterranean Sea. In the southern Teruel Basin

fresh water limestones are also abundant in the Middle Pliocene. In the northern Teruel Basin, however, fresh water limestone deposition terminates at the end of the Gilbert Chron and they are superseded by a red clastic sequence, the Perales Formation. The deposition of these fresh water limestones appears to have been local rather than regional. If local, their significance relative to climatic deterioration that began in the late Miocene (e.g., Messinian Stage) is surely minimal.

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